C U Vivek

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Summary

Senior Performance Engineer with over 10 years of experience architecting Non-Functional Testing (NFT) strategies, distributed system scalability, and enterprise resiliency across Banking, Telecom, and Energy domains. Track record scaling American Express Payment Network Authorization systems to 6,000+ TPS, architecting automated chaos & resiliency validation with Toxiproxy, and engineering autonomous AI performance agents using LangGraph & RAG. Recipient of the American Express Centurion Award 2025.

Experience

American Express, Senior Performance Engineer - Senior Engineer II
Bengaluru, Karnataka, India
June 2021 – present
5 years 5 months

- Architected and deployed an internal AI Performance Agent using **LangGraph**, Vector Databases, and Ollama/Open-Source LLMs to automate performance test execution, multi-dimensional telemetry analysis, and conversational RAG diagnostics of historical defect patterns.
- Spearheaded enterprise scalability and fault-tolerance initiatives for high-throughput payment authorization services.
- Awarded the **Centurion Award 2025**, American Express's premier internal engineering recognition, for technical excellence and impact in system resiliency.
- Led performance engineering for American Express Payment Network Authorization systems, achieving horizontal scalability to **6,000 TPS** via microservices tuning, workload modeling, and thread/heap dump profiling.
- Containerized **Toxiproxy** to simulate network degradation, packet jitter, and latency injection; designed automated chaos engineering experiments across distributed microservices to eliminate resiliency gaps.
- Architected a Shift-Left Performance Testing framework integrated with **GitHub Actions**, enforcing automated throughput and latency validation gates during CI/CD to eliminate post-production defects.
- Executed large-scale distributed load tests using Apache JMeter and Grafana k6 across cloud-native container platforms.
- Monitored and diagnosed distributed transaction traces, connection pools, and JVM garbage collection behavior using Dynatrace and JProfiler.

Amdocs India LLC, Performance Engineer (Amdocs Core R&D Division)
Pune, Maharashtra, India
Nov 2019 – June 2021
1 year 8 months

- Certified monthly drop releases for microservices-based digital applications on **Red Hat OpenShift (Kubernetes)** and **AWS EKS** environments.
- Executed end-to-end performance test monitoring, profiling, and bottleneck analysis across Couchbase, Apache Kafka, Prometheus, Grafana, and AWS services (EC2, RDS, MSK, CloudWatch).
- Conducted Load, Stress, Stability, Scalability, High Availability (HA) Failover, and Chaos Monkey resiliency experiments.
- Analyzed JVM thread and heap dumps with APM tools including Dynatrace, JProfiler, and JVisualVM to size production systems with enterprise architects.

Petrolink Data Services Pvt. Ltd., Performance Engineer
Pune, Maharashtra, India
July 2019 – Oct 2019
4 months

- Formulated Non-Functional Requirements (NFRs) and authored comprehensive Performance Test Plans.
- Engineered end-to-end user traversal test suites using Apache JMeter and profiled server CPU and memory footprints.

Accenture Solutions Pvt. Ltd., Application Development Analyst — Pune, Maharashtra, India
Dec 2016 – July 2019
2 years 8 months

- Developed performance test scripts using HP / Micro Focus LoadRunner across Web (HTTP/HTML) and Web Services protocols.
- Administered StormRunner cloud testing platform, configuring load generator provisioning and debugging execution anomalies.
- Conducted extensive load, stress, and endurance scenarios, analyzing server resource utilization and response times in Dynatrace.

Honors and Awards

Centurion Award 2025 — 2025

- Highest annual engineering recognition at American Express awarded for outstanding technical leadership, resilience engineering, and payment authorization scalability.

Certifications

AI Agents with LangGraph

Hands-On AI: Build a RAG Model from Scratch with Open Source

Microsoft Certified: Azure Fundamentals

AWS Certified Solutions Architect – Associate (SAA)

AWS Certified Cloud Practitioner (CLF)

Education

B.Tech Vellore Institute of Technology (VIT University), Civil Engineering — Vellore, Tamil Nadu, India
July 2012 – June 2016

- Graduated with CGPA: 8.86 / 10

Skills

AI & Autonomous Agents: LangGraph, RAG (Retrieval-Augmented Generation), Vector Databases, Ollama, Open-Source LLMs, Autonomous Telemetry Diagnostics

Performance & Load Testing: Apache JMeter, Grafana k6, Micro Focus LoadRunner, StormRunner, Workload Modeling, Capacity Planning

Observability & APM: Dynatrace (OneAgent), Grafana, Prometheus, JProfiler, JVisualVM, JConsole, Heap & Thread Dump Analysis, GC Tuning

Resiliency & Chaos: Toxiproxy, Chaos Engineering, Chaos Monkey, Latency & Jitter Injection, Failover & High Availability (HA)

Cloud & Containers: Kubernetes, Red Hat OpenShift, AWS (EKS, EC2, RDS, MSK, CloudWatch), Docker

Distributed Data & CI/CD: Apache Kafka, Couchbase, Python, Bash Shell Scripting, GitHub Actions, Jenkins